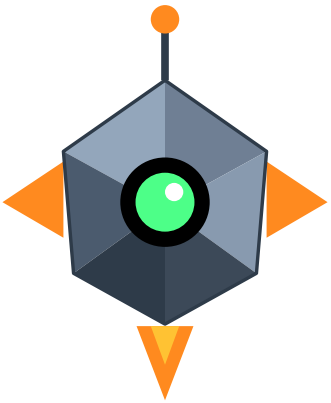
BOOK 5

Reversing the Calculation

Winding a calculation back





MULTIX // HELLO
I am Multix. I ride along in every chapter. Watch my readout: it shows what is owed, what is paid, and what is only a guess.

CHAPTER 1

Many Roads to Twelve



MULTIX // CIRCLE CHECK
A circle holds a piece that belongs together, and circles can sit inside circles. A circle is *ready* when everything needed for its calculation is known and its operation is allowed by the current rules. To **pop** a ready circle, replace it with its value and leave everything outside it alone. To **redraw**, move circles around or split a number, as long as the value stays the same. The full rules are in the introduction.

STORY

Lena's friend Omar left a calculator on her desk with a single number glowing on the screen: **12**. Next to it was a note: *"Guess what I typed."*
Lena stared at the screen. She had no idea where to start.

STEP 1 How many roads lead to 12?

The screen only showed the answer. It did not show what Omar had pressed. So Lena made a list of everything he could have done.

Adding two positive whole numbers to reach 12 (skipping repeats like 5 + 7 and 7 + 5):

1 + 11, 2 + 10, 3 + 9, 4 + 8, 5 + 7, 6 + 6

That is 6 ways already.

Multiplying two positive whole numbers to reach 12:

1 × 12, 2 × 6, 3 × 4

That is 3 more ways.

And there are plenty of other roads: 20 minus 8, 24 divided by 2, 100 minus 88. Lena could keep going for a long time.

The calculator did the work and then forgot how. When a calculation squeezes many different inputs down to one answer, the answer alone cannot tell you which input it came from. Mathematicians say the calculation *throws away information*.

IN CIRCLES

Each road to 12 is a circle that pops to the same number:

Trust check. Imagine Lena guessed "Omar typed 3 times 4" before she had any clue.

TRACE

```
( 5 + 7 )   →  12
( 3 × 4 )   →  12
( 2 × 6 )   →  12
( 20 − 8 )  →  12
```

Omar's calculator showed only the 12. The pop was finished, and the circle that made it was gone. A pop forgets which circle it came from.

STEP 2 Keeping one clue

Omar had typed exactly one operation on two numbers. He felt a little bad, so he gave her a clue: *"The first number I typed was 5, and I added."*

Now Lena had something to work with. If 5 plus something equals 12, she can undo the adding by taking away:

12 minus 5 is 7.

So Omar typed 5 plus 7. One clue was enough to get all the way back.

For the next round, Omar typed a multiplication. The screen showed **12**, and his clue was: *"The first number was 3."*

Lena undid the multiplying by dividing:

12 divided by 3 is 4.

So he had typed 3 times 4. Again, one clue was enough.

IN CIRCLES

Omar's clue turns the mystery into a circle with a hole in it:

TRACE

```
( 5 + ? )  →  12
```

To find the hole, redraw the circle the other way round:

TRACE

```
( 12 − 5 )  →  7
```

For the multiplying round, (3 × ?) pops to 12, and (12 ÷ 3) pops to 4.

STEP 3 The clue that doesn't work

Omar tried a third round, this time allowing himself to use zero. The screen showed **0**, and his clue was: *"The first number was 0, and I multiplied."*

Lena thought about it. 0 times 1 is 0. 0 times 7 is 0. 0 times 1,000 is 0.

Every possible second number gives the same answer. Even with the clue, she could not tell what Omar had typed.

Multiplying by zero is a calculation that throws away *everything* about the second number. No clue about the first number can help, because the damage is done by the zero.

STEP 4 Three ways to remember

There are three ways a machine could keep track of what it did. Each one costs something.

1. **Keep only the answer.** It takes almost no room to store, but you cannot go back. (This is Omar's calculator.)
2. **Keep the answer plus one clue.** It costs a little more room, and sometimes that is enough to go back, as with the 5 and the 3. But not always, as with the zero.
3. **Keep a whole diary.** Write down "start with 5, add 7, got 12." It costs the most room, and it lets you go back every time.

A calculator that has to be tiny and quick keeps only the answer. A bank, which must be able to trace every step, keeps the diary. Neither is wrong. They are trading room against going back. (A diary like this is a trace: one line for each pop.)

MACHINES YOU CAN UNDO

Some machines need no clue at all.

A machine that **adds 7** to whatever you put in can always be undone: take the answer and subtract 7. Different inputs always give different answers, so no two inputs get squished together.

A machine that **rounds to the nearest 10** cannot be undone. Put in 8 and you get 10. Put in 12 and you also get 10. If you see 10 on the screen, you cannot tell which one it was.

The test is simple: **can two different inputs ever give the same output?** If they can, the machine throws information away. If they can't, then any answer the machine really produced has exactly one possible input. Finding that input is not always easy, but you know it is the only one.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the calculator keep?** (The answer.)
- **What did it lose?** (How the answer was made.)
- **What did a clue give back?** (Enough to rebuild the calculation, as long as the clue was the right kind.)

WHAT MAKES THIS CONCLUSION TRUSTWORTHY?

Imagine Lena guessed "Omar typed 3 times 4" before she had any clue. Would that be trustworthy?
No. It would be a guess, and 2 times 6 would have been just as good. Her answer only became trustworthy once she had a clue that ruled out every other road.
(That is the same lesson as the detective in Two Boxes, One Answer: looking at the outside of a machine cannot tell you the inside.)

PRACTICE

1. List every way to make 18 by multiplying two positive whole numbers. Then list every way to make 10 by adding two positive whole numbers, skipping repeats.
2. A machine adds 9 to your number. The screen shows 20. What did you put in? Can this machine always be undone?
3. A machine rounds your number to the nearest 10, with halves rounding up (so 25 becomes 30). The screen shows 30. Which whole numbers could you have put in?
4. A machine multiplies your number by 4. The screen shows 36. What did you put in? Then try the same machine with "multiply by 0." The screen shows 0. Can you tell what went in?
5. A friend says: "Keeping a whole diary is always better than keeping only the answer." Can you think of a time when the diary would be a bad idea? What does it cost?
6. The circle (? + 8) pops to 20. Redraw it the other way round to find the hole.

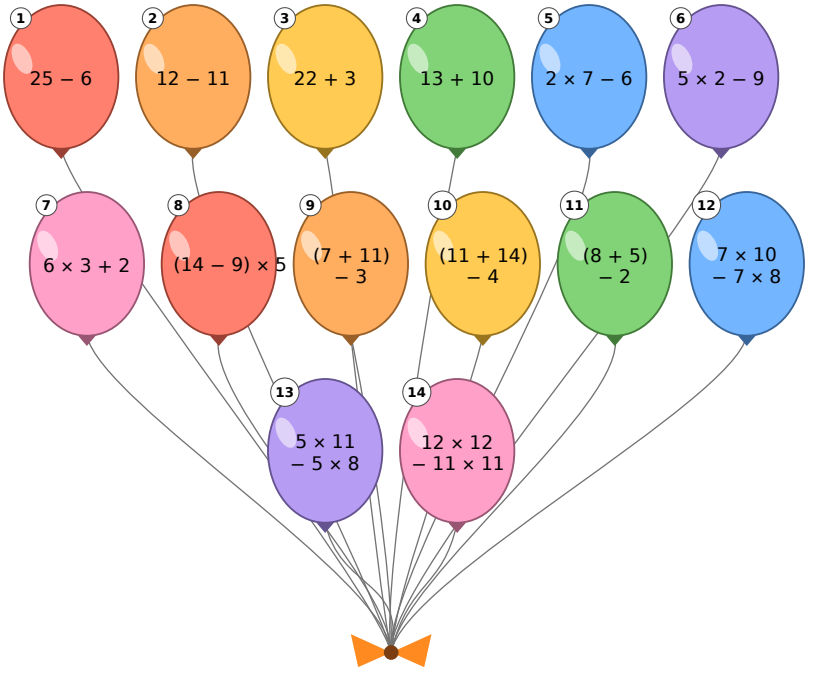
BRIDGE *Bridge to the rest of the series:* In Two Boxes, One Answer, the detective could only see what came out of the slot. Here, the answer is the output, and the detective is trying to work out the input. In A Useful Wrong Answer, Mara's marks were 1 inch apart when they should have been 7/8 inch, and she could still find the right positions because she knew the size of the mistake. In The Order of Discovery, the messy notebook is the diary, and the tidy proof is a short trace that keeps only the steps that matter.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

CHAPTER 2

Socks, Shoes and Buttons That Cannot Be Undone

STORY

Omar built a toy calculator out of cardboard. It had only three buttons: **+3**, **double**, and **round**.
He pressed +3 and double and wrote the final answer on a slip of paper. Lena found the slip. It said **16**.
"What number did you start with?" she asked.
"I'll tell you," said Omar, "but first you have to undo it."

STEP 1 Undo in the opposite order

Omar had pressed: start, then **+3**, then **double**, then the display showed 16.

In circles, with the start number as a mystery *s*:

TRACE

```
( ( s + 3 ) × 2 )  =  16
```

To go backward, Lena undid the last button first.

- The last button was **double**, so she undid it with **halve**: 16 halved is 8.
- Before that was **+3**, so she undid it with **minus 3**: 8 − 3 = 5.

The start was **5**. Check: 5 + 3 = 8, and 8 doubled is 16. ✓

Why did she undo the buttons in the opposite order? Think about socks and shoes. You put on socks first and shoes second. To take them off, you must take off the shoes first. The last thing done is the first thing undone.

STEP 2 Ten doublings

Omar pressed **double** on 1, ten times in a row. Lena wanted to know what appeared on the screen.

TRACE

```
1 → 2 → 4 → 8 → 16 → 32 → 64 → 128 → 256 → 512 → 1,024
```

Trust check. An undo is trustworthy if two different starts can never give the same answer.

Ten doublings of 1 give **1,024**. Then Omar said, “Now I’ll hide the 1,024 and show you only a screen that says 1,024. Which number did I start from?”

Lena pressed **halve** ten times. Each press undoes one double. She walked backward:

TRACE

```
1,024 → 512 → 256 → 128 → 64 → 32 → 16 → 8 → 4 → 2 → 1
```

And she was back at **1**.

Here, double has an undo button that always works. Halve goes back to exactly the number you came from, every time. No information was lost.

STEP 3 The button that cannot be undone

The third button, **round**, rounds a number to the nearest ten, with halves going up. So 23 becomes 20, and 27 becomes 30, and 25 becomes 30.

Omar pressed **round** and the screen said **20**.

“What did I start with?” he asked.

Lena thought. It could have been 20. It could have been 18, or 22, or 15, or 24. All of these round to 20. In fact, any of the ten numbers from 15 to 24 gives 20.

TRACE

```
( round 15 )  →  20
( round 20 )  →  20
( round 24 )  →  20
```

The button **round** has no undo. Once the display says 20, you cannot go back to the start, because many starts lead to the same 20. That is a pop that forgets something it cannot get back.

This is the difference between the two kinds of buttons.

- **Double** gives a different answer for each different start. Each answer points to exactly one start, so the button can be undone.
- **Round** gives the same answer for many starts. One answer points to many starts, so the button cannot be undone.

STEP 4 Keep the diary

What if Omar wants to be able to go back, even with the round

button? Then he should keep a diary of each press, with the numbers before and after.

TRACE

```
start 23   →  round  →  20
```

With the diary, there is a record of the 23. The calculator forgot, but the diary remembered. This is the same idea as the trace in What Must Remain Unresolved.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the double button keep?** (Everything about the start. It can always be undone.)
- **What did the round button keep, and what did it forget?** (It kept roughly how big the number was. It forgot the exact number.)

WHAT MAKES THIS TRUSTWORTHY?

An undo is trustworthy if two different starts can never give the same answer. That is a reason, and you can check it by trying to find a pair of starts that collide. For double there is no such pair. For round there are many. After an undo, always run the undo forward again to check you land on the answer you were given. Lena did: 5 → 8 → 16.

PRACTICE

1. Omar started at some number, pressed **+5**, then **double**, and got 24. What was the start? Show your steps backward.
2. Omar pressed **double** four times starting from 3. What is on the screen? Then undo it.
3. Here is a rule: the screen shows only whether a number is odd or even. Can this be undone? Why or why not?
4. Which numbers round to 30? (Use the same round rule as in the chapter.)
5. A button called **×0** multiplies by zero. Can it be undone? Explain.
6. Omar presses **+3** then **double** then **+3**. In the

screen’s final answer, 25 shows. Find the start.

BRIDGE *Bridge to the rest of the series:* In chapter 1, Lena and Omar saw that many inputs can give one output. Here that idea decides which buttons have an undo. In What Must Remain Unresolved, the trace is the diary that makes round reversible after the fact.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 38 × 42 )
2 redraw  →  ( ( 40 − 2 ) × ( 40 + 2 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 2 × 2 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 2 × 2 )  →  4
6 pop ( 1,600 − 4 )  →  1,596
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 24 × 26 )
2 redraw  →  ( ( 25 − 1 ) × ( 25 + 1 ) )
3 redraw  →  ( ( 25 × 25 ) − ( 1 × 1 ) )
4 pop ( 25 × 25 )  →  625
5 pop ( 1 × 1 )  →  2
6 pop ( 625 − 2 )  →  623
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 46 × 54 )
2 redraw  →  ( ( 50 − 4 ) × ( 50 + 4 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 4 × ( 4 + 1 ) ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 4 × ( 4 + 1 ) )  →  20
6 pop ( 2,500 − 20 )  →  2,480
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 3

Clues That Shrink the Pile

STORY

Omar's round button had a problem. When the screen said 20, there were ten possible starts: every number from 15 to 24. Lena called it **the pile**.
"I can't undo the button," she said. "But maybe I can shrink the pile with clues."

STEP 1 One clue at a time

Omar agreed to give clues about his secret start number. Each clue had to be true.

The pile after the screen said 20:

TRACE

```
15  16  17  18  19  20  21  22  23  24
```

Clue 1: "My number is odd." Lena crossed out the even ones.

TRACE

```
15  17  19  21  23
```

Clue 2: "My number is a multiple of 3." That means it is in the 3 times table. Lena crossed out what was not.

TRACE

```
15  21
```

Clue 3: "Its two digits add up to 6." For 15, 1 + 5 = 6. For 21, 2 + 1 = 3.

TRACE

```
15
```

The secret start was **15**.

Lena checked by pressing the round button on 15: it rounds up to 20, because halves go up. The screen said 20. ✓

Trust check. The answer is trustworthy when you can check it two ways.

STEP 2 How clues work

Each clue is a rule. A number stays in the pile only if it passes the rule. A good clue cuts the pile a lot. A weak clue cuts almost nothing.

- “My number is less than 100” crosses out nothing from 15 to 24. It was a useless clue.
- “My number is odd” crossed out about half.
- A clue like “My number is bigger than 30” crosses out the entire pile. That means one of the clues must be wrong, or Omar made a mistake.

So a good set of clues is **true** (nothing contradicts) and **enough** (only one number is left).

STEP 3 How many clues are enough?

It depends on the pile and the clues. With the pile 15 to 24:

- The two clues “odd” and “multiple of 3” left two numbers, 15 and 21. Not enough.
- Adding “digits add up to 6” left one. Enough.

With a different third clue, such as “less than 20,” only 15 would have been left. With “bigger than 20,” only 21 would have been left. So the clues that finish the job can be different, as long as they split the pile the right way.

STEP 4 The diary shortcut

Lena realised she could skip all the clues if Omar had kept a diary, as in chapter 2:

TRACE

```
start 15  →  round  →  20
```

A diary gives the answer straight away. Clues are for when there is no diary: they bring a number back by shrinking the pile.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the round button keep?** (A pile of ten possible starts.)
- **What did each clue keep?** (Only the numbers that fit it.)

WHAT MAKES THIS TRUSTWORTHY?

The answer is trustworthy when you can check it two ways. First, the number passes **every clue**. Second, running the button forward on the number gives the screen you started with. Lena did both for 15.
The pile also tells you when a set of clues is not enough, because more than one number is still standing. Do not pick one. Ask for another clue.

PRACTICE

1. The screen says 30 after the round button. List the pile. (Use halves go up.)
2. Omar adds: “My number is even.” Which numbers are left?
3. He adds: “It is a multiple of 4.” Which are left? Then: “Its digits add up to 5.” What is the secret number?
4. Is “my number is less than 100” a useful clue for the pile in problem 1? Why or why not?
5. The screen says 20. Omar gives two clues: “even” and “bigger than 25.” What happens?

BRIDGE *Bridge to the rest of the series:* In chapter 2, the round button had no undo because many starts gave one screen. Here that many-to-one pile is shrunk by clues. In What Must Remain Unresolved, the clue $x + y = 15$ does the same job: it narrows what is unknown without naming it.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **1,599**. Leave the others blank. What appears?

1575 + 16	1574 + 13	1584 + 13	1582 + 17	1609 − 10	1582 + 18	1573 + 14	1602 + 4
14 × 114	1563 + 28	1615 − 16	1573 + 26	1561 + 38	1562 + 37	58 × 58 − 42 × 42	1594 + 17
53 × 30	1620 − 21	1614 − 15	1562 + 37	1569 + 30	68 × 68 − 55 × 55	1627 − 28	1584 + 9
1620 − 21	1637 − 38	1614 − 15	1621 − 22	1593 + 6	1614 − 15	68 × 68 − 55 × 55	1594 + 5
1623 − 24	1575 + 24	1591 + 8	1603 − 4	1567 + 32	1616 − 17	1586 + 13	1628 − 29
1616 − 17	1576 + 23	1566 + 33	134 × 12	1576 + 25	1586 + 13	1607 − 8	1622 − 23
1595 + 4	1577 + 22	1629 − 30	1584 + 4	50 × 50 − 30 × 30	1569 + 30	1622 − 23	13 × 123
1576 + 23	1566 + 33	1604 − 5	1576 + 15	1589 + 13	1594 + 5	1632 − 33	1579 + 20



MULTIX//SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 4

Buttons on a Clock

STORY

Omar built a calculator with a clock face instead of a screen. It had the numbers 0 to 11, where 12 o'clock is called 0. When a number goes past 11, it wraps around, just like a clock.
So 9 + 5 on this clock is 14, which wraps to **2**.

STEP 1 A button that can be undone

The first button is **+5**. Starting from 3, it gives 8. Starting from 9, it gives 14, which wraps to 2.

Can it be undone? Omar looked for a button that sends every answer back. Adding 5 and then adding 7 makes 12, and 12 wraps back to 0. So **+7 undoes +5**.

TRACE

```
9  →  +5  →  2
2  →  +7  →  9
```

Because each start gives a different answer, +5 can always be undone.

STEP 2 The ×5 button

Omar added a **×5** button: multiply by 5, then wrap. Here are all twelve starts:

TRACE

```
start:   0  1  2  3  4  5  6  7  8  9  10 11
×5:      0  5  10 3  8  1  6  11 4  9  2  7
```

Look at the second row. Every number from 0 to 11 appears exactly once. So each answer points to one start, and **×5 can be undone**.

What undoes it? Try ×5 a second time on the answers: 5 × 5 = 25, which wraps to 1. So pressing ×5 twice brings you back to where you began. A button that undoes itself!

Trust check. The list of all twelve starts is a complete test, because the clock has only twelve numbers.

TRACE

```
3  →  ×5  →  15 wraps to 3     (3 is a lucky one)
2  →  ×5  →  10  →  ×5  →  50 wraps to 2
```

STEP 3 The ×2 button

Now the **×2** button.

TRACE

```
start:   0  1  2  3  4  5  6  7  8  9  10 11
×2:      0  2  4  6  8  10 0  2  4  6  8  10
```

Starts 0 and 6 both give 0. Starts 1 and 7 both give 2. Every answer appears twice, and the odd numbers 1, 3, 5, 7, 9, 11 never appear as answers at all. So **×2 cannot be undone**: if the clock says 4, the start might have been 2 or 8.

It is the same thing that happened with the round button in chapter 2. Many starts, one answer.

STEP 4 Which multiply buttons are safe?

Omar tried ×3, ×4 and ×7 on the 12-clock too.

- **×3**: 0, 3, 6, 9, 0, 3 … lots of repeats. Cannot be undone.
- **×4**: 0, 4, 8, 0, 4, 8 … cannot be undone.
- **×7**: 0, 7, 2, 9, 4, 11, 6, 1, 8, 3, 10, 5. Every number once. **Can be undone**, by pressing ×7 again, because 7 × 7 = 49, which wraps to 1.

Omar noticed that the safe buttons were 1, 5, 7 and 11. The unsafe ones shared a factor with 12: 2, 3, 4, 6, 8, 9, 10. This is a pattern from tries, and the reason is beyond this book. Omar marked it *only seen so far*, as Kofi would.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What does a button that can be undone keep?** (Every detail of the start.)
- **What does a button that cannot be undone lose?** (Which of several starts it was.)

WHAT MAKES THIS TRUSTWORTHY?

The list of all twelve starts is a complete test, because the clock has only twelve numbers. That is a rare case where testing every start is possible, and so a table is a real proof here.
A button that undoes itself is checked by pressing it twice and landing back at the start.

PRACTICE

1. On the 12-clock, what is 8 + 7? Which button undoes +7?
2. Press ×5 on 4. What do you get? Press ×5 again. What do you get?
3. Start at 2. Press +5, then ×5. What is the answer? Undo it step by step.
4. Why can ×6 not be undone on the 12-clock? Give two starts with the same answer.
5. List ×7 for the starts 1, 2 and 3.

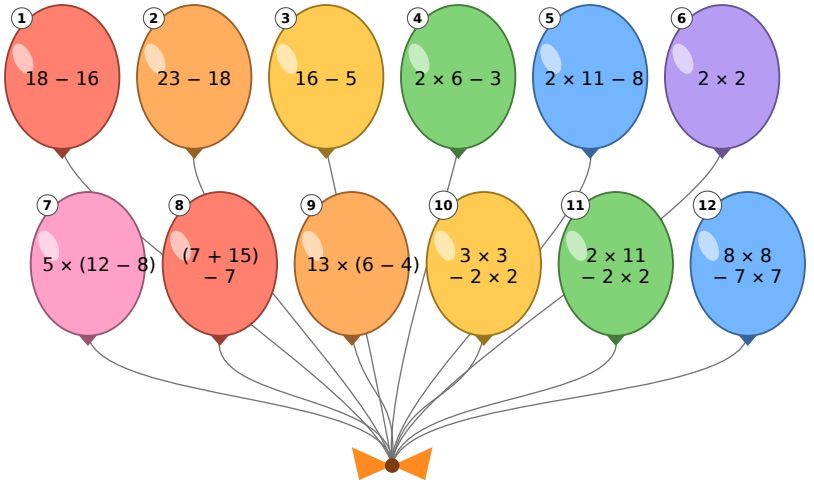
BRIDGE *Bridge to the rest of the series:* In chapter 2, whether a button can be undone depended on whether two starts give one answer. On a small clock, that can be tested for every start. In Two Boxes, One Answer, a table of every case is the only kind of test that settles anything.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

											!
1	2	3	4	5	6	7	8	9	10	11	12

REVERSING THE CALCULATION

CHAPTER 5

Which Lines Can Be Undone

STORY

Lena kept a notebook for 15 × 12. It had six lines.

TRACE

```
( 15 × 12 )
redraw  →  ( ( 15 × 10 ) + ( 15 × 2 ) )
pop ( 15 × 10 )  →  150
pop ( 15 × 2 )  →  30
( 150 + 30 )
pop  →  180
```

Omar covered up everything except the last line: **180**. "Can you tell me what the problem was?" he asked. Lena could not. Many problems give 180: 18 × 10, 20 × 9, 90 × 2, 100 + 80. The bare answer had forgotten the road.

STEP 1 Undo with the notebook

Omar uncovered the lines, one by one, from the bottom.

Lena could undo each line by looking at what was written.

- Last line, *pop → 180*: the line above it says what was popped, (150 + 30). So 180 goes back to (150 + 30).
- Before that, *pop (15 × 2) → 30*: the line records (15 × 2). So 30 goes back to (15 × 2).
- Then *pop (15 × 10) → 150*, which goes back to (15 × 10).
- Then the redraw. A redraw keeps the value, so undoing it means redrawing the other way: ((15 × 10) + (15 × 2)) back to (15 × 12).

She walked all the way back to the start.

STEP 2 Which lines can be undone without help?

Lena sorted the lines into two kinds.

- **A redraw can always be undone**, even without a record. It kept the value and only rearranged the circles, so you can rearrange back. Nothing was forgotten.

Trust check. A notebook is a trustworthy way to go backward when every pop line records what was popped.

- **A pop cannot be undone without a record.** The pop *pop (15 × 10) → 150* replaced a circle with its value. If you only see 150, you cannot tell it came from 15 × 10 and not from 30 × 5 or 75 × 2.

So the lines that need the notebook are the pops. The notebook records what was inside each circle, so the pop can be read backward.

TRACE

```
150  could be:  ( 15 × 10 )   ( 30 × 5 )   ( 75 × 2 )   ( 100 + 50 )
```

STEP 3 A line that changes nothing

Suppose the notebook also had a refusal line, like Sade's in The Boundary of a Step.

TRACE

```
refuse ( 15 × 11 )  →  not allowed: the problem asks for exactly 15 × 12
```

Nothing was changed by a refusal, so there is nothing to undo. But the line still says something: it tells the reader why a move was not taken.

So three kinds of lines behave differently.

Line	Can it be undone?	Why
redraw	yes, always	the value was kept
pop	only with the record	the circle's insides were forgotten
refusal	nothing to undo	no change was made

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What does the notebook keep that the answer 180 does not?** (What each pop replaced.)
- **Which line forgets the most?** (A pop.)

WHAT MAKES THIS TRUSTWORTHY?

A notebook is a trustworthy way to go backward when every pop line records what was popped. If a line only says "→ 150" without saying what was inside the circle, that pop cannot be undone, and neither can anything after it.

PRACTICE

1. Redraw (15 × 12) into ((15 × 10) + (15 × 2)). Show the redraw that undoes it.
2. Give three different circles that pop to 24.
3. A notebook line says only "→ 24". Can you undo it? What would you need?
4. Which kind of line is easy to undo even if you lose the notebook: a redraw or a pop? Explain.
5. Omar hides the notebook and says the final answer is 180. Give two problems that fit.

BRIDGE *Bridge to the rest of the series:* In chapter 4, a button on the clock could be undone when each answer came from one start. Here the same question is asked of lines in a notebook. In What Must Remain Unresolved, the notebook is also the thing that keeps the note to self faithful.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 37 × 43 )
2 redraw  →  ( ( 40 − 3 ) × ( 40 + 3 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 3 × ( 3 + 1 ) ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 3 × ( 3 + 1 ) )  →  12
6 pop ( 1,600 − 12 )  →  1,588
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 19 × 21 )
2 redraw  →  ( ( 20 − 1 ) × ( 20 + 1 ) )
3 redraw  →  ( ( 20 × 20 ) − ( 1 × 1 ) )
4 pop ( 20 × 20 )  →  400
5 pop ( 1 × 1 )  →  2
6 pop ( 400 − 2 )  →  398
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 64 × 76 )
2 redraw  →  ( ( 70 − 6 ) × ( 70 + 6 ) )
3 redraw  →  ( ( 70 × 70 ) − ( 6 × 6 ) )
4 pop ( 70 × 70 )  →  4,900
5 pop ( 6 × 6 )  →  36
6 pop ( 4,900 − 36 )  →  4,864
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 6

The Think-of-a-Number Trick

STORY

Omar did a trick at lunch. He said:
Think of a number. Add 5. Double it. Subtract 10. Halve it. Now tell me your answer, and I will tell you your starting number.
Lena thought of 7 and followed the steps.

TRACE

```
7  →  +5  →  12
12  →  ×2  →  24
24  →  −10  →  14
14  →  ÷2  →  7
```

“7,” said Lena. “That’s my starting number!”
“Of course it is,” said Omar. “The trick always gives you your own number back.”

STEP 1 Why it works

Lena tried it again with 20.

TRACE

```
20 → 25 → 50 → 40 → 20
```

Back to 20 again. She could not tell why. She drew circles for a number n she did not know:

TRACE

```
( ( ( n + 5 ) × 2 ) − 10 )
redraw  →  ( ( ( n × 2 ) + ( 5 × 2 ) ) − 10 )
pop ( 5 × 2 )  →  10
( ( ( n × 2 ) + 10 ) − 10 )
redraw  →  ( ( n × 2 ) + ( 10 − 10 ) )
pop  →  ( n × 2 )
```

After the first three steps, the number n has been doubled. The +10 and −10 cancelled. The last step halves it, which undoes the doubling. That is why n comes back.

Lena noticed the buttons come in pairs of undos, as in chapter 2: ×2 and ÷2 undo each other, and +10 and −10 undo each other.

STEP 2 A trick that always gives the same number

Trust check. The reason is the circles with n.

Omar showed a second trick:

Think of a number. Double it. Add 6. Halve it. Subtract the number you started with. The answer is always 3.

Lena tried 4: double is 8, add 6 is 14, halve is 7, subtract 4 is **3**. She tried 10: 20, 26, 13, minus 10 is **3**.

Why?

TRACE

```
( ( ( n × 2 ) + 6 ) ÷ 2 )
redraw  →  ( n + 3 )
( ( n + 3 ) − n )
redraw  →  ( 3 + ( n − n ) )
pop  →  3
```

The n vanishes. Everything about the starting number is gone, and only the 3 is left. That is why the answer is always 3.

STEP 3 Making your own trick

To make a trick, choose steps that undo each other, or steps that make the unknown cancel.

Lena built one: *Think of a number. Triple it. Add 9. Divide by 3. Subtract your number. The answer is always 3.* She tested it with 5: 15, 24, 8, minus 5 is 3. ✓

Test it with 1: 3, 12, 4, minus 1 is 3. ✓

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the first trick keep?** (The starting number, all the way through.)
- **What did the second trick lose?** (The starting number: it cancels out completely.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is the circles with n. Every step is drawn with the unknown still inside, so we are not testing numbers one by one: we are showing that the trick works for all of them.
Testing with 7, 20 and 4 is evidence. The circles are the reason.

PRACTICE

1. Try the first trick with 12. What do you get at each step?
2. Try the second trick with 15. What is the answer?
3. Check the triple-and-add-9 trick with 8.
4. Make a trick that always ends in 5. (Hint: double, add 10, halve, subtract the number.)
5. Why would the trick fail if Omar said “subtract 12” instead of “subtract 10” in the first trick?

BRIDGE *Bridge to the rest of the series:* In chapter 5, a redraw was the one line that could always be undone, and here the redraws are what prove the trick. In What Must Remain Unresolved, working with n unknown is exactly the way to show something is true for every number.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **1,764**. Leave the others blank. What appears?

65 × 27	1755 + 16	1745 + 15	24 × 60 + 18 × 18	3 × 81 + 39 × 39	1745 + 17	1743 + 24	93 × 19
1757 + 14	27 × 65	1757 + 7	32 × 52 + 10 × 10	1802 − 38	58 × 58 − 40 × 40	1738 + 16	1735 + 22
17 × 104	23 × 61 + 19 × 19	1745 + 19	16 × 68 + 26 × 26	1750 + 14	1767 − 3	1755 + 9	1725 + 28
1771 − 7	14 × 126	1741 + 23	4 × 441	1737 + 27	6 × 78 + 36 × 36	19 × 65 + 23 × 23	1743 + 21
1770 − 6	84 × 21	15 × 69 + 27 × 27	1758 + 6	4 × 80 + 38 × 38	1736 + 28	34 × 50 + 8 × 8	1754 + 10
1778 − 14	196 × 9	70 × 70 − 56 × 56	1756 + 11	31 × 57	1761 + 3	1782 − 18	30 × 54 + 12 × 12
6 × 78 + 36 × 36	1790 − 26	22 × 62 + 20 × 20	1762 + 10	1726 + 26	4 × 80 + 38 × 38	9 × 75 + 33 × 33	18 × 66 + 24 × 24
33 × 51 + 9 × 9	1794 − 30	1796 − 32	1736 + 22	1766 + 7	126 × 14	1732 + 32	14 × 126



MULTIX//SCAN

Some boxes look close and are not. Check the corner on each one.

REVERSING THE CALCULATION

CHAPTER 7

Undoing a Percentage

STORY

A coat is on sale for 20% off. The sale price is **48** dollars. What was the original price?
Lena's first idea was to add 20% back.

STEP 1 The tempting wrong answer

20% of 48 is 9.6. Add it back:

TRACE

```
( 48 + 9.6 )  →  57.6
```

Lena checked it. She took 20% off 57.6 and got 57.6 − 11.52 = 46.08. That is not 48. So 57.6 is wrong.

Why? The 20% was taken off the **original** price, not off 48. Twenty percent of the original is a bigger number than twenty percent of the sale price. Adding the wrong 20% back gives the wrong answer.

STEP 2 Write the percent as a button

Taking 20% off leaves 80% of the price. 80% is four fifths. So the "20% off" button is:

TRACE

```
( ( original × 4 ) ÷ 5 )  =  48
```

Now Lena undoes the buttons in the opposite order, as in chapter 2.

- Undo ÷5: 48 × 5 = 240.
- Undo ×4: 240 ÷ 4 = 60.

The original price was **60** dollars. Check: 60 × 4 = 240, 240 ÷ 5 = 48. ✓

Trust check. The reason is that "20% off" is exactly "multiply by 4, divide by 5", and every button of that kind can be undone.

STEP 3 A rise instead of a fall

A game costs more now because the price went up by 25%. The new price is 75 dollars. What was it before?

A rise of 25% leaves 125% of the old price, which is five quarters.

TRACE

```
( ( original × 5 ) ÷ 4 )  =  75
```

- Undo ÷4: 75 × 4 = 300.
- Undo ×5: 300 ÷ 5 = 60.

The old price was **60** dollars. Check: 60 + 25% of 60 is 60 + 15 = 75. ✓

STEP 4 Why the wrong way is tempting

"20% off" and "20% on" look like opposites. They are not. Take 20% off 60 and you get 48. Add 20% to 48 and you get 57.6, not 60.

TRACE

```
( 60 − 12 )  →  48
( 48 + 9.6 )  →  57.6
```

A percent is a share of whatever number it is applied to. It takes different amounts from different numbers. So the button for "20% off" cannot be undone by "20% on". The undo button is a different percent: to get from 48 back to 60, you add 25% of 48.

TRACE

```
( 48 + 12 )  →  60       (12 is 25% of 48)
```

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What does the undo keep?** (The original price, found exactly.)
- **What did the tempting method get wrong?** (Which number the percent was taken of.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is that "20% off" is exactly "multiply by 4, divide by 5", and every button of that kind can be undone. The check is to apply the original percent to the answer and see if you land on the sale price.

PRACTICE

1. After 10% off, a shirt costs 45 dollars. What was the original? (90% is nine tenths.)
2. After a rise of 50%, a ticket costs 90 dollars. What was the old price? (150% is three halves.)
3. After 25% off, a hat costs 30 dollars. What was the original?
4. After 20% off, a bag costs 36 dollars. What was the original?
5. Why is 30 + 25% of 30 not the answer to problem 3?

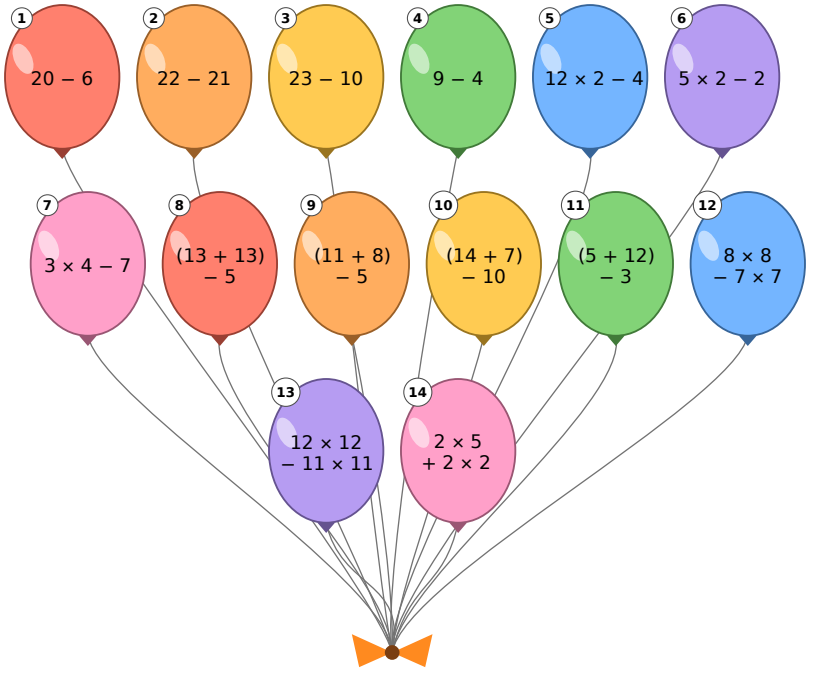
BRIDGE *Bridge to the rest of the series:* In chapter 2, buttons were undone in the opposite order. Here a percent is written as two buttons so it can be undone the same way. In A Useful Wrong Answer, a share has to say what it is a share of. *Next in this book:* Chapter 8: Winding 2,496 Back to 48 and 52.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

REVERSING THE CALCULATION

CHAPTER 8

Winding 2,496 Back to 48 and 52

STORY

Noor found a card with only an answer on it: **2,496**. Under it, a clue: *"Two numbers, 4 apart, were multiplied to make this."*
Which two numbers? She decided to run the case file backward.

STEP 1 Read the trace from the bottom

Here is the trace of 48 × 52 that Noor already knew:

TRACE

```
( 48 × 52 )
redraw  →  ( ( 50 − 2 ) × ( 50 + 2 ) )
redraw  →  ( ( 50 × 50 ) − ( 2 × 2 ) )
pop ( 50 × 50 )  →  2,500
pop ( 2 × 2 )  →  4
pop ( 2,500 − 4 )  →  2,496
```

To undo it, she goes up from the bottom, one line at a time, and undoes each line with its opposite. Undoing is done in the **reverse order**, like taking off shoes before socks.

- The last line took 4 away from 2,500. The opposite is to add 4: 2,496 + 4 = **2,500**.
- The line before that made 2,500 from 50 × 50. The opposite is to ask, *what number times itself makes 2,500?* That is **50**.
- The first line split the numbers around 50. The clue says the numbers are 4 apart, so each is 2 from the middle: **48 and 52**.

Noor checks it forward: 48 × 52 = 2,496. ✓

STEP 2 Why the clue was needed

A pop forgets how its value was made. The number 2,496 does not remember that it came from 2,500 − 4, or from 48 × 52. A trace remembers, but the card had no trace.

Without the clue, there are many pairs that multiply to 2,496:

Trust check. Noor trusts her answer because she ran it forward and got 2,496 again.

TRACE

```
26 × 96     39 × 64     48 × 52     24 × 104
```

Every one of these is a correct answer to "what multiplies to 2,496?". The clue *4 apart* picks out just one: 48 and 52 are 4 apart, but 39 and 64 are 25 apart, and the other pairs are even further.

So winding back needed two things: the answer, and a clue about what was thrown away.

STEP 3 Why this clue works

The clue "4 apart" means each number is 2 from the middle. The corner is $2 \times 2 = 4$, so the product is a square minus 4. That is why adding 4 gives a square, and why the square's root is the middle.

If the clue had been "6 apart", each number would be 3 from the middle, and the corner would be $3 \times 3 = 9$. Then Noor would add 9, not 4.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the answer 2,496 keep?** (The product, but not the pair that made it.)
- **What did the clue bring back?** (The distance between the numbers, which is enough to find the pair.)

WHAT MAKES THIS TRUSTWORTHY?

Noor trusts her answer because she ran it forward and got 2,496 again. Undoing and redoing is a check that catches slips. She also knows why the clue was enough: it fixed the corner, and the corner fixed the pair.

PRACTICE

1. The product is **891** and the two numbers are 6 apart. Find them.
2. The product is **2,499** and the two numbers are 2 apart. Find them.
3. The product is **9,996** and the two numbers are 4 apart. Find them.

4. The product is **2,491** and the two numbers are 6 apart. Find them.

5. Noor says, “2,496 alone tells me the two numbers.” Show that she is wrong with two different pairs.

BRIDGE *Bridge to the rest of the series:* In What Must Remain Unresolved, the clue about the distance is what turns an unknown into a number. In A Useful Wrong Answer, the same corner was the debt on an estimate.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 38 × 42 )
2 redraw  →  ( ( 40 − 2 ) × ( 40 + 2 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 2 × 2 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 2 × 2 )  →  4
6 pop ( 1,600 − 4 )  →  1,596
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 24 × 26 )
2 redraw  →  ( ( 25 − 1 ) × ( 25 + 1 ) )
3 redraw  →  ( ( 25 × 25 ) − ( 1 × 1 ) )
4 pop ( 25 × 25 )  →  625
5 pop ( 1 × 1 )  →  2
6 pop ( 625 − 2 )  →  623
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 46 × 54 )
2 redraw  →  ( ( 50 − 4 ) × ( 50 + 4 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 4 × ( 4 + 1 ) ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 4 × ( 4 + 1 ) )  →  20
6 pop ( 2,500 − 20 )  →  2,480
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

Answers (turn the book around)

Chapter 1

1. Multiplying to 18: **1 × 18, 2 × 9, 3 × 6** (three ways). Adding to 10: **1 + 9, 2 + 8, 3 + 7, 4 + 6, 5 + 5** (five ways).
2. 20 minus 9 is **11**. Yes: different inputs always give different answers, so you can always subtract 9 to go back.
3. Any whole number from **25 to 34**. That is ten different inputs, so the machine cannot be undone.
4. 36 divided by 4 is **9**. Multiplying by 4 can always be undone. But "multiply by 0" always gives 0, so you cannot tell what went in.
5. The diary costs room, and it can cost speed too. A tiny machine, or one that does millions of steps, might not have space to write down every one. Also, if the steps are private, a diary lets other people read them. Whether to keep the diary depends on how much you need to go back, and how much room you can spare.
6. (20 − 8) pops to **12**, so the hole is 12.

Balloon Message.

1.	25 − 6	= 19	S
2.	12 − 11	= 1	A
3.	22 + 3	= 25	Y
4.	13 + 10	= 23	W
5.	2 × 7 − 6	= 8	H
6.	5 × 2 − 9	= 1	A
7.	6 × 3 + 2	= 20	T
8.	(14 − 9) × 5	= 25	Y
9.	(7 + 11) − 3	= 15	O
10.	(11 + 14) − 4	= 21	U
11.	(8 + 5) − 2	= 11	K
12.	7 × 10 − 7 × 8	= 14	N
13.	5 × 11 − 5 × 8	= 15	O
14.	12 × 12 − 11 × 11	= 23	W

Message: **SAY WHAT YOU KNOW!**

Chapter 2

1. Undo the double first: 24 halved is 12. Then undo +5: 12 − 5 = **7**. Check: 7 + 5 = 12, and 12 doubled is 24. ✓
2. 3 → 6 → 12 → 24 → 48, so the screen shows **48**. Undoing: 48 → 24 → 12 → 6 → 3.
3. No. Many numbers are odd, and many are even, so the screen cannot tell you which one was the start.
4. The numbers from **25 to 34**: 25 rounds up to 30, and 34 rounds down to 30.
5. No. Any start, multiplied by 0, gives 0. So the 0 on the screen could have come from any number.
6. Undo the last +3: 25 − 3 = 22. Undo the double: 22 ÷ 2 = 11. Undo the first +3: 11 − 3 = **8**. Check: 8 + 3 = 11, double is 22, plus 3 is 25. ✓

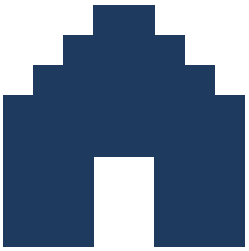
Spot the Slip.
Notebook A: **perfect**. 38 × 42 = 1,596.
Notebook B: slip on line **5**. 1 × 1 is 1. The answer should be 624.
Notebook C: slip on line **3**. The corner is 4 × 4, not 4 × (4 + 1). The answer should be 2,484.

Answers (turn the book around)

Chapter 3

1. The pile is **25, 26, 27, 28, 29, 30, 31, 32, 33, 34**.
2. **26, 28, 30, 32, 34**.
3. Multiples of 4 among those: **28 and 32**. Digits add to 5: 3 + 2 = 5 for 32, and 2 + 8 = 10 for 28, so the number is **32**. Check: round 32 gives 30. ✓
4. No. Every number in the pile is already below 100, so the clue crosses out nothing.
5. The pile for 20 is 15 to 24. "Even" leaves 16, 18, 20, 22, 24. "Bigger than 25" crosses out all of them. The clues cannot all be true together, so somebody made a mistake.

Color by Answer (1,599). The shaded boxes make a house.



Chapter 4

1. 8 + 7 = 15, which wraps to **3**. The button **+5** undoes +7, because 7 + 5 = 12.
2. 4 × 5 = 20, which wraps to **8**. Then 8 × 5 = 40, which wraps to **4**. Back at the start.
3. 2 + 5 = 7, then 7 × 5 = 35, which wraps to **11**. To undo, press ×5 on 11: 55 wraps to 7. Then +7: 14 wraps to **2**. ✓
4. Starts 1 and 3 both give 6 (1 × 6 = 6 and 3 × 6 = 18, which wraps to 6). So the answer 6 does not say which start it came from.
5. 1 × 7 = **7**, 2 × 7 = 14 wraps to **2**, 3 × 7 = 21 wraps to **9**.

Balloon Message.

1.	18 − 16	= 2	B
2.	23 − 18	= 5	E
3.	16 − 5	= 11	K
4.	2 × 6 − 3	= 9	I
5.	2 × 11 − 8	= 14	N
6.	2 × 2	= 4	D
7.	5 × (12 − 8)	= 20	T
8.	(7 + 15) − 7	= 15	O
9.	13 × (6 − 4)	= 26	Z
10.	3 × 3 − 2 × 2	= 5	E
11.	2 × 11 − 2 × 2	= 18	R
12.	8 × 8 − 7 × 7	= 15	O

Message: **BE KIND TO ZERO!**

Answers (turn the book around)

Chapter 5

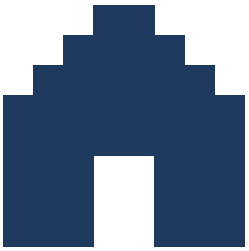
1. The redraw back is ((15 × 10) + (15 × 2)) → (15 × 12). It collects the two lots into one.
2. For example (4 × 6), (3 × 8), (12 + 12). Others work too.
3. No. You would need the line that says what was popped, such as (4 × 6) → 24.
4. A **redraw**. It only rearranged the circles and kept the value, so rearranging back always works. A pop forgot what was inside.
5. For example (15 × 12) and (20 × 9). Any circle that pops to 180 fits.

Spot the Slip.
Notebook A: slip on line **3**. The corner is 3 × 3, not 3 × (3 + 1). The answer should be 1,591.
Notebook B: slip on line **5**. 1 × 1 is 1. The answer should be 399.
Notebook C: **perfect**. 64 × 76 = 4,864.

Chapter 6

1. 12 → 17 → 34 → 24 → **12**.
2. 15 → 30 → 36 → 18 → 18 − 15 = **3**.
3. 8 → 24 → 33 → 11 → 11 − 8 = **3**. ✓
4. Double, add 10, halve, subtract the number. For 6: 12, 22, 11, minus 6 is **5**. In circles, ((n × 2) + 10) ÷ 2 = n + 5, and subtracting n leaves 5.
5. The +10 and −12 would not cancel exactly. You would get back n − 1, not n.

Color by Answer (1,764). The shaded boxes make a house.



Answers (turn the book around)

Chapter 7

1. ((original × 9) ÷ 10) = 45. Undo ÷10: 450. Undo ×9: 450 ÷ 9 = **50** dollars.
2. ((old × 3) ÷ 2) = 90. Undo ÷2: 180. Undo ×3: 180 ÷ 3 = **60** dollars.
3. 75% is three quarters. ((original × 3) ÷ 4) = 30. Undo ÷4: 120. Undo ×3: **40** dollars.
4. 80% is four fifths. Undo ÷5: 180. Undo ×4: 180 ÷ 4 = **45** dollars.
5. 30 + 25% of 30 = 37.50, but the 25% was taken off the original 40, not off 30. Twenty-five percent of 40 is 10, and 30 + 10 = 40.

Balloon Message.

1.	20 − 6	= 14	N
2.	22 − 21	= 1	A
3.	23 − 10	= 13	M
4.	9 − 4	= 5	E
5.	12 × 2 − 4	= 20	T
6.	5 × 2 − 2	= 8	H
7.	3 × 4 − 7	= 5	E
8.	(13 + 13) − 5	= 21	U
9.	(11 + 8) − 5	= 14	N
10.	(14 + 7) − 10	= 11	K
11.	(5 + 12) − 3	= 14	N
12.	8 × 8 − 7 × 7	= 15	O
13.	12 × 12 − 11 × 11	= 23	W
14.	2 × 5 + 2 × 2	= 14	N

Message: **NAME THE UNKNOWN!**

Chapter 8

1. Each is 3 from the middle, so add 3 × 3 = 9: 891 + 9 = 900. The middle is 30. The numbers are **27 and 33**. (27 × 33 = 891 ✓)
2. Each is 1 from the middle, so add 1: 2,500. The middle is 50. The numbers are **49 and 51**.
3. Add 4: 10,000. The middle is 100. The numbers are **98 and 102**.
4. Add 9: 2,500. The middle is 50. The numbers are **47 and 53**. (47 × 53 = 2,491 ✓)
5. For example, 39 × 64 = 2,496 and 26 × 96 = 2,496. Both match, and neither is 48 × 52.

Spot the Slip.
Notebook A: **perfect**. 38 × 42 = 1,596.
Notebook B: slip on line **5**. 1 × 1 is 1. The answer should be 624.
Notebook C: slip on line **3**. The corner is 4 × 4, not 4 × (4 + 1). The answer should be 2,484.